Diamond chips, from wafer to qubit

ADAMAS: an open, tested framework for diamond electronics and room-temperature quantum processors

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Aleksander Norman

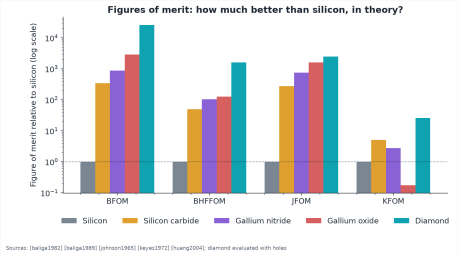
Every number on these slides is generated by the package (`make talk`)

California State University, Dominguez Hills · github.com/Normansrule/adamas-diamond-framework

Why diamond at all?

- Power-switch figure of merit: 25,777× silicon
- Heat removal: 14.7× silicon
- A defect, the nitrogen-vacancy (NV) center, is a qubit at room temperature

But: boron dopants are 0.37 eV deep (under 1% active at 300 K), wafers only reached 3 inches recently, and logic is p-channel only.



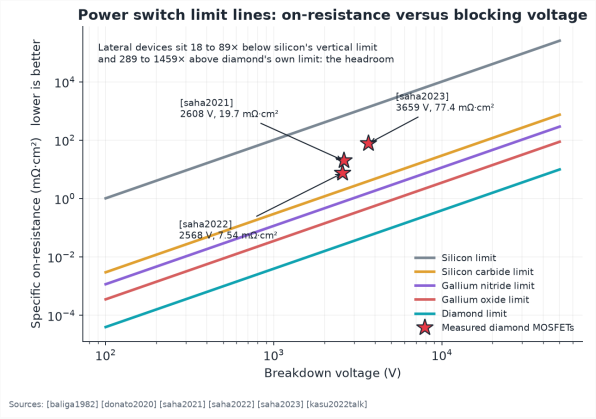
What ADAMAS is

One connected, executable chain, each link checked against published data or an independent tool:

materials → process → circuits → qubits → error correction → machine size → experiments

- 583 references, 500 verified against Crossref
- 53 figures regenerated from code
- Validation matrix: 18 checks, 17 pass
- Process traveler, PDK, placed 4-bit processor
- Circuit-level error correction (Stim, PyMatching)
- Experiments from US$100, course, 22 web pages

Power switches: diamond wins when hot



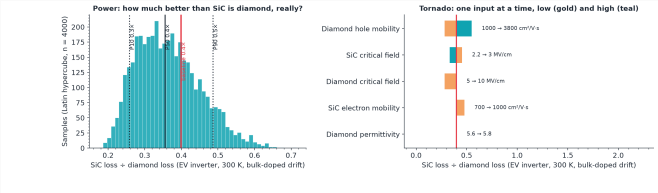
Best measured: 3659 V; 2022 lateral device about 89× below silicon's vertical limit.

Bulk-doped drift: only 0.50% of the boron ionized at 300 K.

W per switch	300 K	523 K
Silicon	209.5	–
SiC	29.7	58.0
GaN	20.6	–
Diamond, bulk	74.6	29.8

800 V, 300 A, 20 kHz EV inverter

But how sure is that? Uncertainty analysis

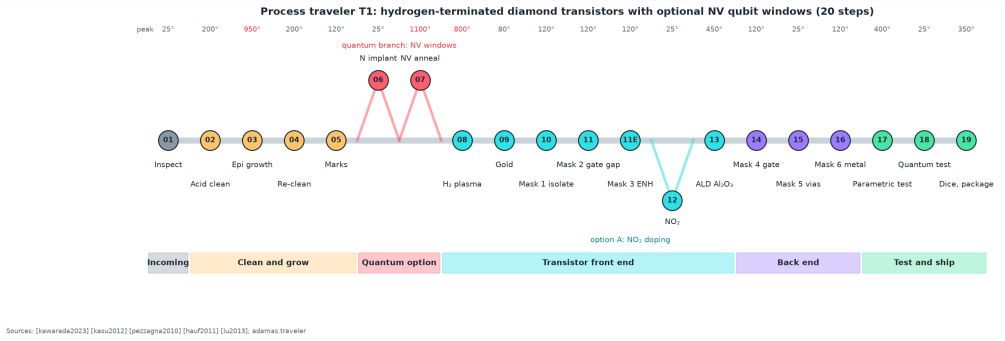


Diamond's critical field (5 to 10 MV/cm) and hole mobility (1,000 to 3,800, baseline 2,000) are not settled.

SiC ÷ diamond loss at 300 K: median 0.36, P10 to P90 0.26 to 0.49; at 523 K: median 1.7×.

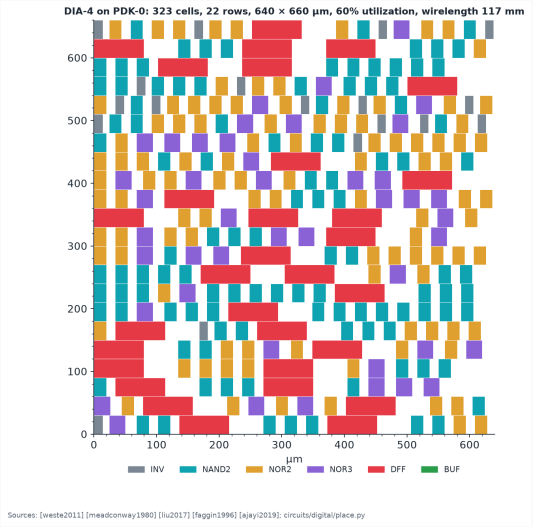
Mobility explains 0.64 of the spread, the field 0.17: measure mobility first (a Hall setup, about US$3k).

How you would actually build it



The channel is made by surface chemistry (hydrogen termination), gold protects it, one oxygen plasma isolates every device, and all heat is spent before the first metal. Qubit wafers: hydrogen below 500 °C only (hot plasma darkens NV centers).

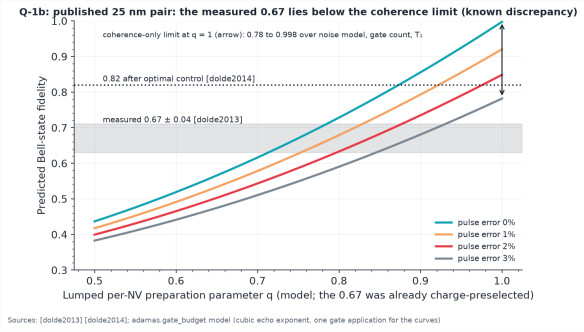
A processor on diamond



Sources: [weste2011] [meadconway1980] [liu2017] [faggin1996] [ajayi2019]; circuits/digital/place.py

- PDK-0: layers, rules, GDS monitor die, standard cells, LEF, timed Liberty
- DIA-4 (4-bit, CALL/RET) synthesizes to 323 cells, about 1 mm^2 with pads
- Browser emulator matches the Verilog on random programs; ring oscillator within 10% of SPICE

Two-qubit gates: an honest discrepancy

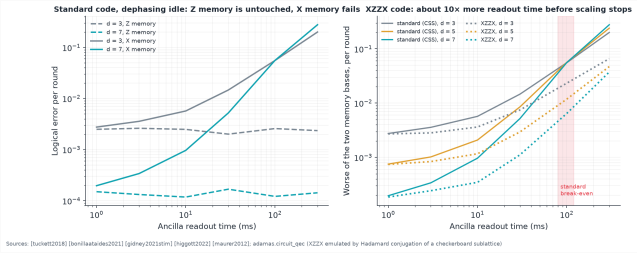


Published room-temperature pair (25 nm, 25 µs gate): coherence alone allows 0.78 to 0.998.

Measured 0.67 (already charge-preselected) lies below the whole range.

The earlier "preparation" explanation is withdrawn; the gap is a known discrepancy. 96% at 9.5 nm has since been shown [joas2025].

Error correction when readout is slow



Sources: [tuckett2018] [bonillaataides2021] [gidney2021stim] [higgott2022] [maurer2012]; adamas.circuit_qec (XZZX emulated by Hadamard conjugation of a checkerboard sublattice)

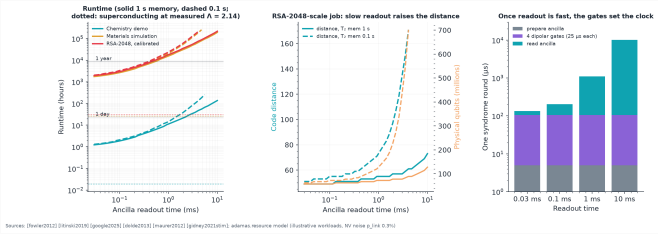
Readout takes about 1 ms while data qubits idle.

$\Lambda = 4.03$ (0.1 ms), 3.56 (1 ms), 2.64 (10 ms).

Scaling stops near 100 ms; the bias-tailored XZZX code pushes that to about 1000 ms.

Native circuits, bias on CZ only (CNOT ≤ 5): threshold 0.93% $\rightarrow$ 1.81% at $\eta = 100$ (standard code: 0.86%).

How big, how slow



Sources: [fowler2012] [litinski2019] [google2025] [dekle2013] [maurer2012] [gidney2021stim]; adamas resource model (illustrative workloads; NV noise p_{link} 0.3%).

RSA-2048, calibrated to the published 8 h: distance 55, 73 M qubits on a 5.5 mm die.

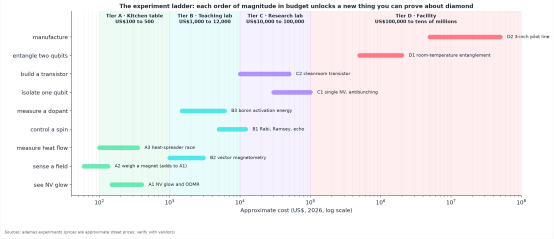
Runtime 2.1 years at 1 ms readout (0.3 at 0.1 ms), P10 to P90 0.4 to 5.3 years; superconducting at today's Λ: 30 hours.

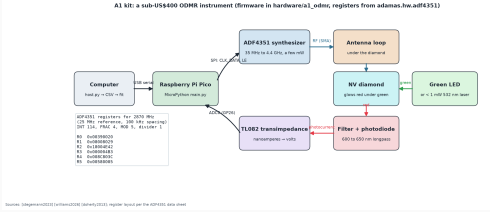
Assumes a 25 µs, 0.3% inter-cell link (not yet demonstrated).

Native circuits: biasing the gates moves the standard code from 135 to 173 M qubits and XZZX from 142 to 58 M.

Area is free; time is not.

Anyone can start: experiments from US$100





A1 kit: Pico firmware drives an ADF4351 through the 2.87 GHz resonance; data go straight into the fitter.

Takeaways

1. Diamond power switches win when hot (1.9× less loss than SiC at 523 K) and a bulk-doped switch loses at room temperature: incomplete ionization matters.
2. Room-temperature NV gates measure below what coherence allows, and the gap is unexplained; addressing qubits 10–30 nm apart is the manufacturing problem.
3. A room-temperature quantum computer would be small but slow; readout speed and bias-tailored codes are the levers.
4. Everything is open, tested, and regenerable: `pytest`, `make figures paper talk`.

github.com/Normansrule/adamas-diamond-framework
normansrule.github.io/adamas-diamond-framework